

Python Built-in Functions

A Beginner's Guide

Python provides many built-in functions that are readily available without importing any libraries. These functions simplify programming by performing common tasks.

1. print()

Purpose: Displays output on the screen.

Syntax

```
print(value)
```

Example

```
name = "John"
print(name)
```

Output

```
John
```

2. input()

Purpose: Accepts input from the user.

Syntax

```
input("Enter value: ")
```

Example

```
name = input("Enter your name: ")
print("Hello", name)
```

Output

```
Enter your name: Alice
Hello Alice
```

3. type()

Purpose: Returns the data type of an object.

Example

```
x = 25  
print(type(x))
```

Output

```
<class 'int'>
```

4. len()

Purpose: Returns the number of elements in an object.

Example

```
text = "Python"  
print(len(text))
```

Output

```
6
```

5. int()

Purpose: Converts a value into an integer.

Example

```
age = "22"  
age = int(age)  
print(age)  
print(type(age))
```

Output

```
22  
<class 'int'>
```

6. float()

Purpose: Converts a value into a floating-point number.

Example

```
number = "10"  
number = float(number)  
print(number)
```

Output

10.0

7. str()

Purpose: Converts a value into a string.

Example

```
number = 100
text = str(number)
print(text)
print(type(text))
```

Output

100
<class 'str'>

8. bool()

Purpose: Converts a value into a Boolean.

Example

```
print(bool(1))
print(bool(0))
```

Output

True
False

9. abs()

Purpose: Returns the absolute value.

Example

```
print(abs(-45))
```

Output

45

10. round()

Purpose: Rounds a number to the specified number of decimal places.

Example

```
pi = 3.14159  
print(round(pi, 2))
```

Output

```
3.14
```

11. max()

Purpose: Returns the largest value.

Example

```
numbers = [10, 50, 25, 80]  
print(max(numbers))
```

Output

```
80
```

12. min()

Purpose: Returns the smallest value.

Example

```
numbers = [10, 50, 25, 80]  
print(min(numbers))
```

Output

```
10
```

13. sum()

Purpose: Returns the sum of all values.

Example

```
numbers = [10, 20, 30]  
print(sum(numbers))
```

Output

14. sorted()

Purpose: Returns a sorted list.

Example

```
numbers = [40, 10, 25, 5]
print(sorted(numbers))
```

Output

```
[5, 10, 25, 40]
```

15. range()

Purpose: Generates a sequence of numbers.

Example

```
for i in range(5):
    print(i)
```

Output

```
0
1
2
3
4
```

16. enumerate()

Purpose: Returns both the index and the value while iterating.

Example

```
fruits = ["Apple", "Orange", "Mango"]
for index, fruit in enumerate(fruits):
    print(index, fruit)
```

Output

```
0 Apple
```

1 Orange

2 Mango

17. zip()

Purpose: Combines multiple iterables.

Example

```
names = ["Alice", "Bob"]
marks = [90, 85]

for name, mark in zip(names, marks):
    print(name, mark)
```

Output

```
Alice 90
Bob 85
```

18. list()

Purpose: Creates a list.

Example

```
text = "Python"
letters = list(text)
print(letters)
```

Output

```
['P', 'y', 't', 'h', 'o', 'n']
```

19. tuple()

Purpose: Creates a tuple.

Example

```
numbers = [1, 2, 3]
t = tuple(numbers)
print(t)
```

Output

(1, 2, 3)

20. set()

Purpose: Creates a set and removes duplicate values.

Example

```
numbers = [1, 2, 2, 3, 3, 4]
print(set(numbers))
```

Output

```
{1, 2, 3, 4}
```

21. dict()

Purpose: Creates a dictionary.

Example

```
student = dict(name="John", age=22)
print(student)
```

Output

```
{'name': 'John', 'age': 22}
```

22. id()

Purpose: Returns the memory address (identity) of an object.

Example

```
x = 100

print(id(x))
```

Output

```
140736812345680 (The value will differ on each computer.)
```

23. help()

Purpose: Displays documentation for a function or module.

Example

```
help(print)
```

Output

Help on built-in function print...

24. dir()

Purpose: Lists all available methods and attributes.

Example

```
text = "Python"  
print(dir(text))
```

Output (Partial)

```
['capitalize', 'casefold', 'center', 'count', ...]
```

25. isinstance()

Purpose: Checks whether an object belongs to a specific data type.

Example

```
x = 25  
print(isinstance(x, int))  
print(isinstance(x, str))
```

Output

```
True  
False
```

26. pow()

Purpose: Raises a number to a given power.

Example

```
print(pow(2, 5))
```

Output

```
32
```

27. divmod()

Purpose: Returns the quotient and remainder.

Example

```
print(divmod(17, 5))
```

Output

```
(3, 2)
```

28. any()

Purpose: Returns True if at least one element is True.

Example

```
values = [False, False, True]  
print(any(values))
```

Output

```
True
```

29. all()

Purpose: Returns True only if all elements are True.

Example

```
values = [True, True, False]  
print(all(values))
```

Output

```
False
```

30. eval()

Purpose: Evaluates a string as a Python expression.

Example

```
expression = "10 + 20"  
print(eval(expression))
```

Output

```
30
```

Note: Avoid using eval() with user input because it can execute arbitrary Python code.

Function	Description
print()	Displays output
input()	Reads user input
type()	Returns data type
len()	Returns length
int()	Converts to integer
float()	Converts to float
str()	Converts to string
bool()	Converts to Boolean
abs()	Absolute value
round()	Rounds numbers
max()	Largest value
min()	Smallest value
sum()	Sum of values
sorted()	Sorts values
range()	Generates sequences
enumerate()	Returns index and value
zip()	Combines iterables
list()	Creates a list
tuple()	Creates a tuple
set()	Creates a set
dict()	Creates a dictionary
id()	Returns object identity
help()	Displays documentation
dir()	Lists methods
isinstance()	Checks data type
pow()	Calculates powers
divmod()	Quotient and remainder
any()	Checks if any value is True
all()	Checks if all values are True
eval()	Evaluates expressions